Economic Factors

* Economic / Variable load factors

Avg load (P_{avg}) :

$$P_{avg} = \frac{\text{Area under load curve (kWhrs)}}{\text{Total time in Hrs}}$$

Plant load factor (Plf)

$$Plf = \frac{P_{avg}}{P_{max}} = \frac{P_{avg} \times T}{P_{max} \times T}$$ Ideal Plf = 1, practical Plf < 1

$$= \frac{\text{Area under load curve (kWhrs)}}{\text{Rectangular Area corresponding to } P_{max}}$$

* Plant capacity factor (Pcf)

$$Pcf = \frac{P_{avg}}{P_c}$$ P_c = plant capacity

Ideal Pcf = 1, practical Pcf < 1

$$= \frac{\text{Energy produced in total hrs}}{\text{Energy able to produce as per } P_c \text{ in total hrs}}$$

* Plant Usage Factor (Puf)

$$Puf = \frac{\text{Energy produced in used hrs}}{\text{Energy able to produce as per } P_c \text{ in used hrs}}$$

Ideal Puf = 1, practical Puf < 1

* Utilization factor (Uf)

$$U_f = \frac{P_{max}}{P_c}$$

Ideal Uf = 1, practical Uf < 1

* Reserve Capacity (R_c)

- R_c = plant capacity - Max. Demand

$= P_c - P_{max}$

$= \frac{P_{max}(Plf - Pcf)}{Pcf}$

* Demand factor (D_f)

$$D_f = \frac{P_{max}}{\text{Sum of connected load}}$$

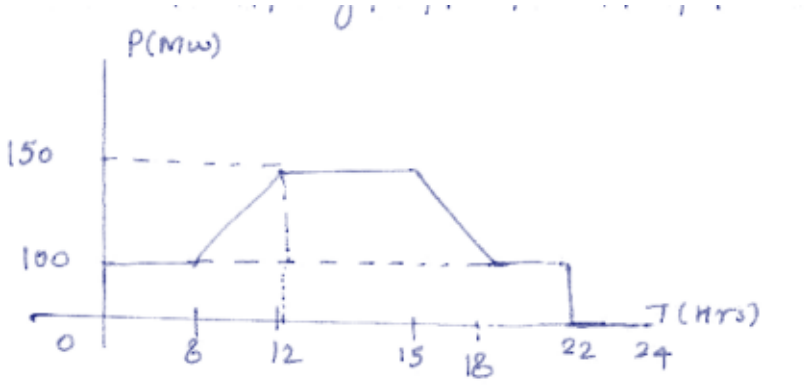
Ideal $Df = 1$, practical $Df < 1$

* Diversity factor (Div_f)

$$Div_f = \frac{\text{Sum of individual Max. Demand}}{P_{max} \text{ of station}}$$

Practical : $Divf > 1$

eg: A Gen station has installed capacity of 200 MW and daily load curve is shown below. Find P_c, P_{max}, P_{avg}, Plf, Pcf, Puf, Uf, Rc ?



Daily load curve: 100 MW to 150 MW (8-12 hrs, 15-18 hrs), 22-24 hrs no load

$P_c = 200$ MW, $P_{max} = 150$ MW

$$P_{avg} = 800 + \frac{1}{2} \times 4 \times 50 + 100 \times 4 + \ldots$$

$$P_{avg} = \frac{(100 \times 22) + \frac{1}{2} \times 4(50) + 50 \times 3 + \frac{1}{2} \times 3 \times 50 + (0 \times 2)}{24}$$

$$P_{avg} = \frac{2525}{24} = 105.20833 \text{ MW}$$

$$PIf = \frac{105.20833}{150} = 0.70138$$

$$Pcf = \frac{105.20833}{200} = 0.526$$

$$U_f = \frac{150}{200} = 0.75$$

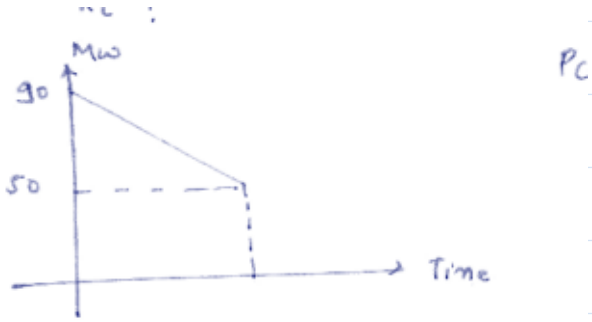
$$R_c = 200 - 150 = 50 \text{ MW}$$

Puf =

$$PIf = \frac{2525}{150 \times 24} = 0.70138, \quad Pcf = \frac{2525}{200 \times 24} = 0.526$$

$$Puf = \frac{2525}{200 \times 22} = 0.57$$

W.B Q.22] R_c = ?



Load curve: 90 MW falling to 50 MW

Pcf = 70% = 0.7, P_{max} = 90

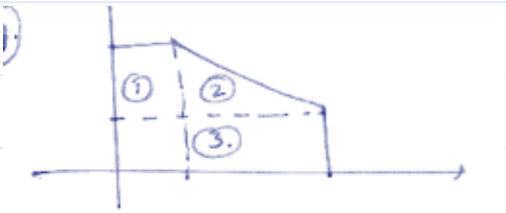
$$PIf = \frac{\frac{1}{2} \times 100 \times 40 + 50 \times 100}{90 \times 100} = 0.77$$

$$R_c = \frac{P_{max}(PIf - Pcf)}{Pcf} = \frac{90(0.77 - 0.7)}{0.7} = 10\ MW$$

Q.23] load factor = 0.6 , P_c = 720 MWh , P_{max} = ?

$$PIf = \frac{P_{avg}}{P_{max}} = \frac{720}{24 \times 0.6} = 50\ MW$$

Q.24]



Load curve with areas 1, 2, 3

$$PIf = \frac{90 \times 1000 + \frac{1}{2} \times 60 \times 90 + 10 \times 8760}{100 \times 8760} = 0.6013 \Rightarrow 60.13\%$$

Q.25]

$$Pcf = \frac{10 \times 12 + 5 \times 12}{15 \times 24} = 0.5$$

Q.26] P_{max} = 15000 kW , PIf = 50/100 , Pcf = 40/100

$$R_c = \frac{15000(0.5 - 0.4)}{0.4} = 3750$$

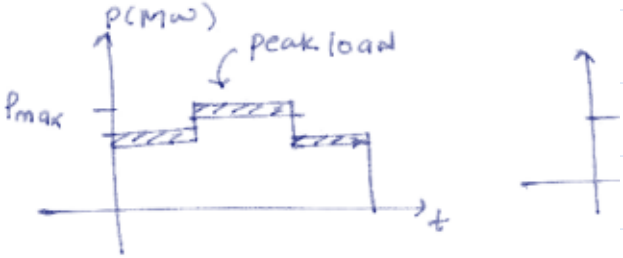
* In a practical power system Diversity factor > 1 and PIf equal to 1 is preferred, which will reduce the cost of electrical energy. It is due to changes in the P_{max} value.

Methods of improving Plf and Diversity factor

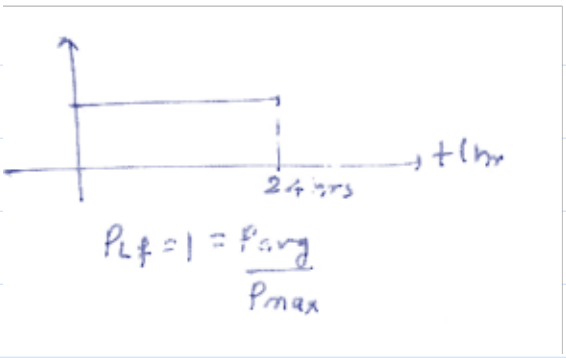
(1) Providing subsidy to the industries and agricultural areas for running their electrical equipment during off peak load period.

(2) Effective utilization of solar energy so that energy can be utilized depending on the light source available timings.

(3) Constant load curve (Plf = 1): base load generating stations able to supply entire load demand so that overall cost is reduced.



Peak load curve with Pmax



Constant load curve, Plf = 1 = Pavg / Pmax (24 hrs)